I. Dataset Summary (IDD Detection)

TRAIN Split	VAL Split	TEST Split
4033 Images | 4033 Labels	2633 Images | 2633 Labels	1047 Images | 858 Labels

View Detailed Dataset Analysis

📊 EDA for the `Train` Split

📁 Data Integrity Check

Total images found: 4033

Total label files found: 4033

📈 Dataset Statistics

Objects with no annotations: `0` images

Object Distribution by Class:

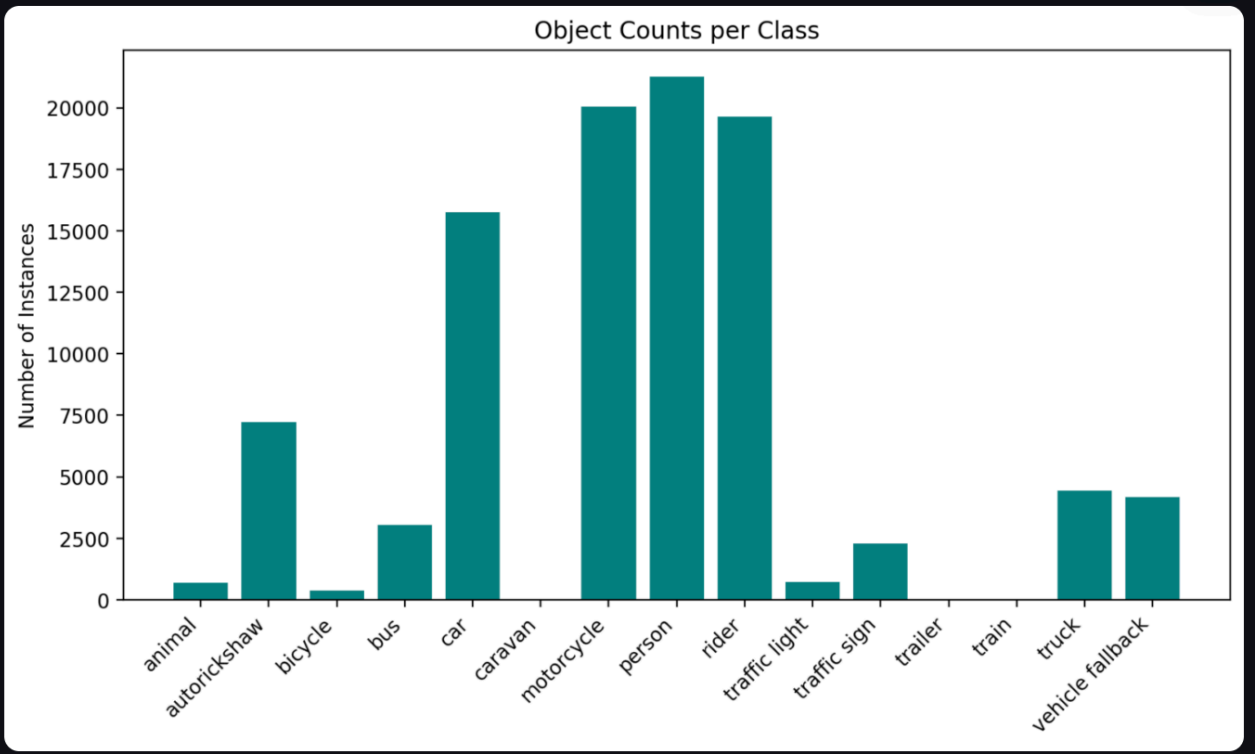
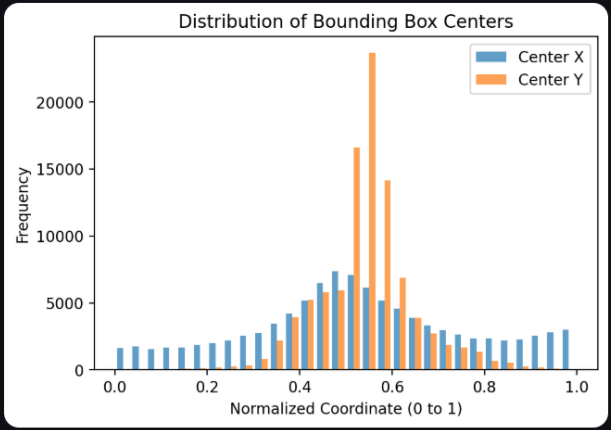


Image and Bounding Box Analysis

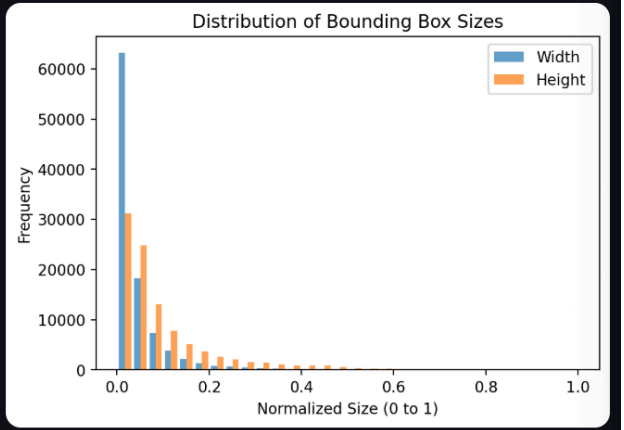
Image Dimensions: Max W: `1921` px, Max H: `1080` px. Avg W: `1905` px, Avg H: `1077` px.

Bounding Box Center Distribution



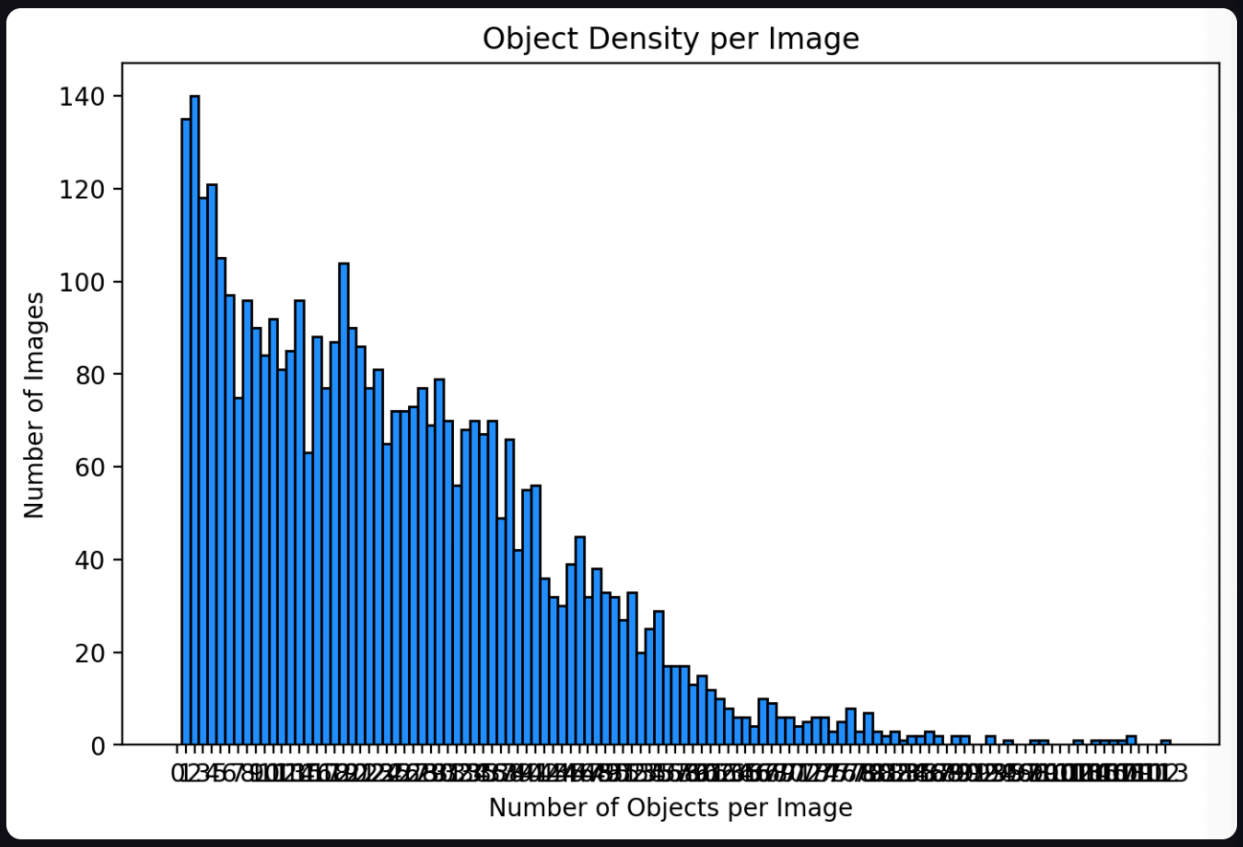
This plot reveals if objects are biased towards specific regions of the images.

Bounding Box Size Distribution



This plot helps understand the typical scale of objects in the dataset (small objects dominate if bars are near 0).

📦 Object Density Analysis



This histogram shows how many objects are typically present in each image, indicating dataset sparsity or density.

📊 EDA for the `Val` Split

📁 Data Integrity Check

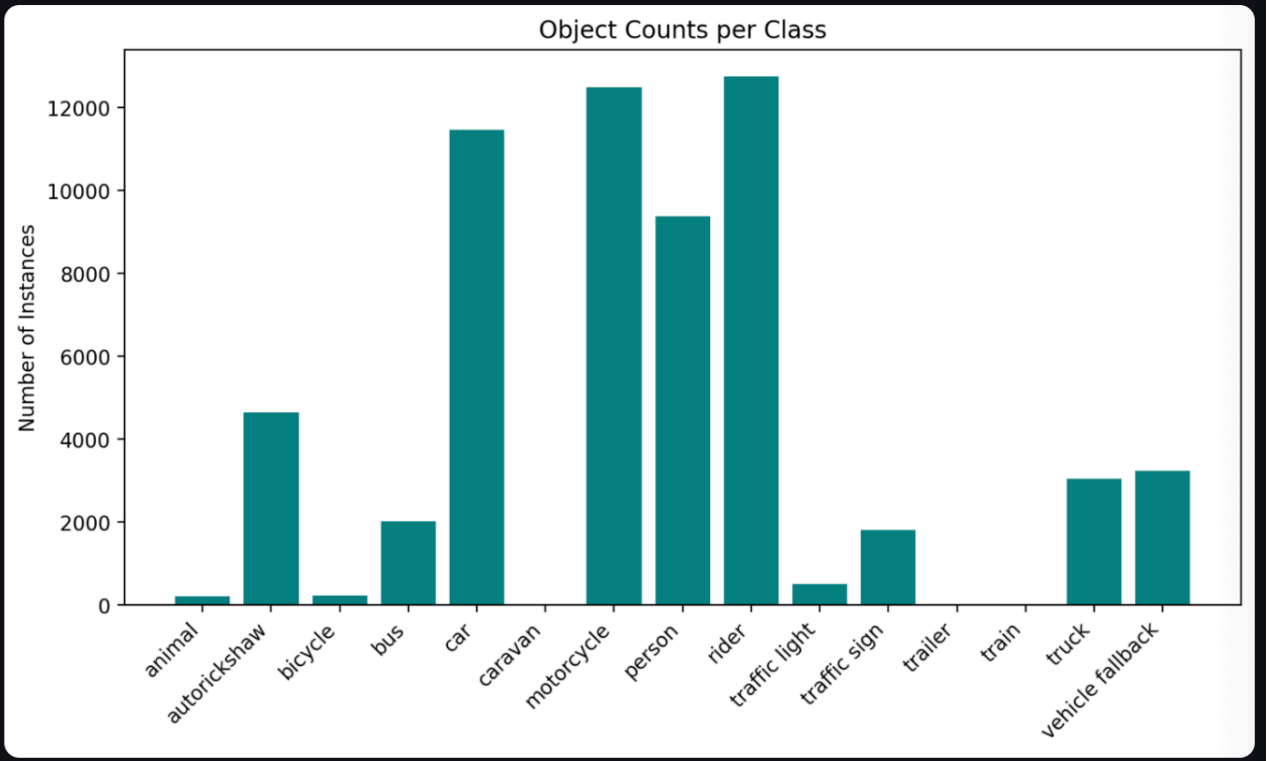
Total images found: 2633

Total label files found: 2633

📈 Dataset Statistics

Objects with no annotations: `0` images

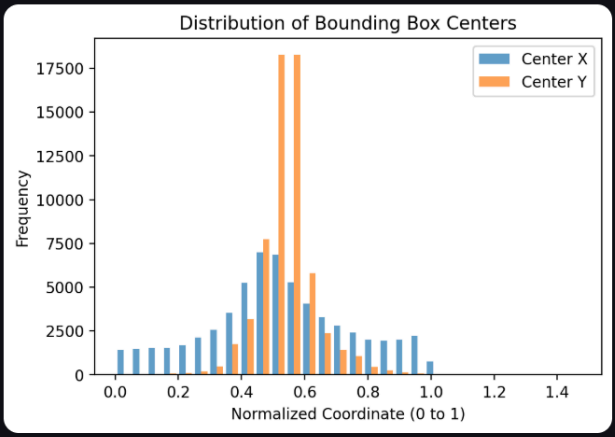
Object Distribution by Class:



🖼️ Image and Bounding Box Analysis

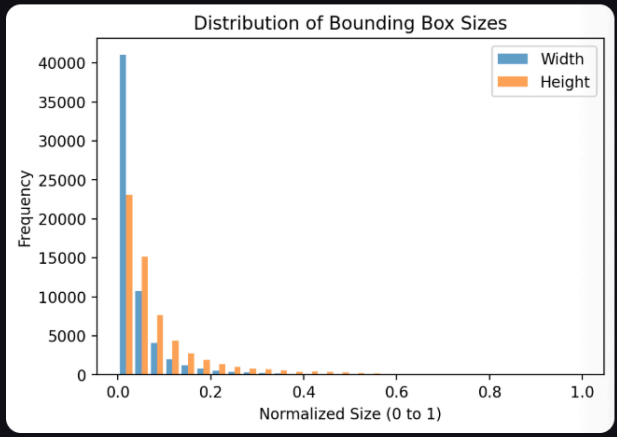
Image Dimensions: Max W: `1921` px, Max H: `1080` px. Avg W: `1810` px, Avg H: `1026` px.

Bounding Box Center Distribution



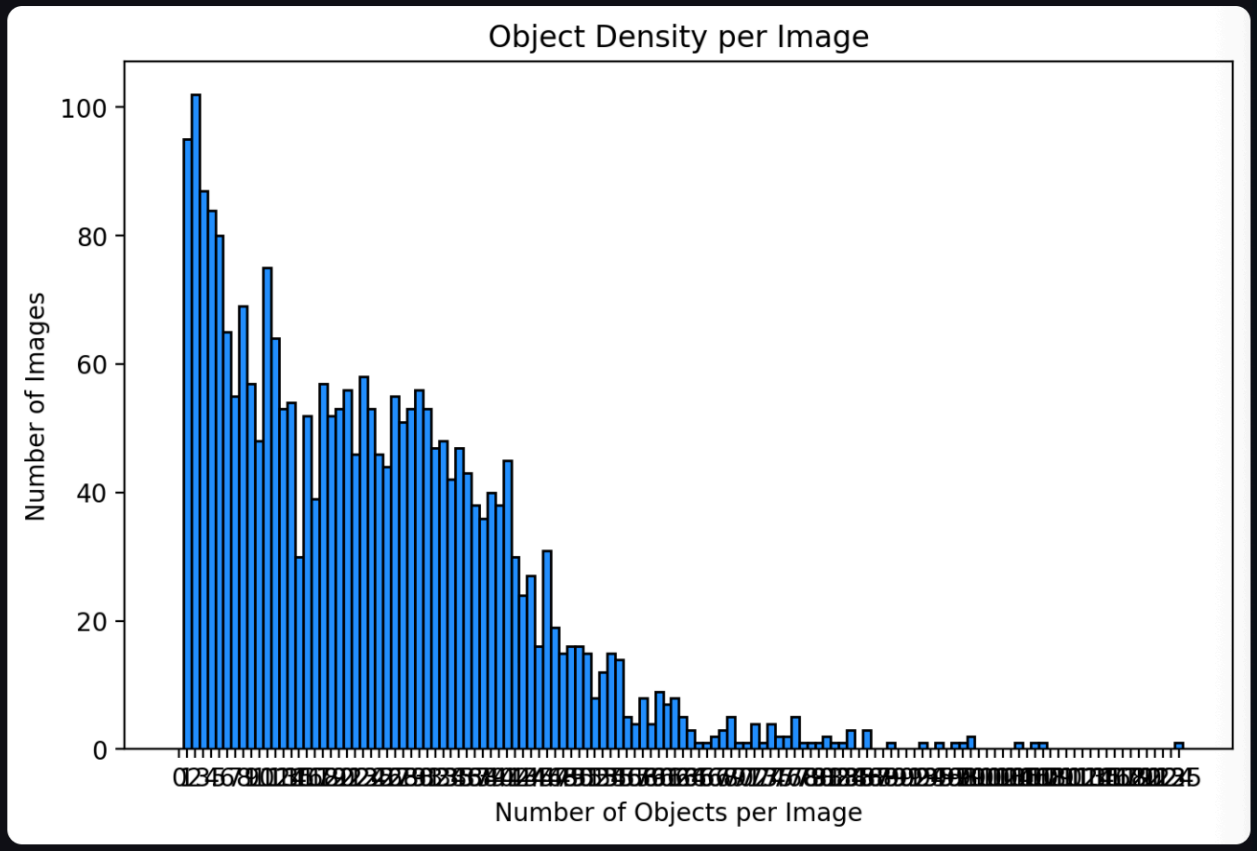
This plot reveals if objects are biased towards specific regions of the images.

Bounding Box Size Distribution



This plot helps understand the typical scale of objects in the dataset (small objects dominate if bars are near 0).

📦 Object Density Analysis



This histogram shows how many objects are typically present in each image, indicating dataset sparsity or density.

📊 EDA for the `Test` Split

📁 Data Integrity Check

Total images found: 1047

Total label files found: 858

⚠️ **189 images have no corresponding labels.** They will be excluded from analysis.

📈 Dataset Statistics

Objects with no annotations: `0` images

Object Distribution by Class:

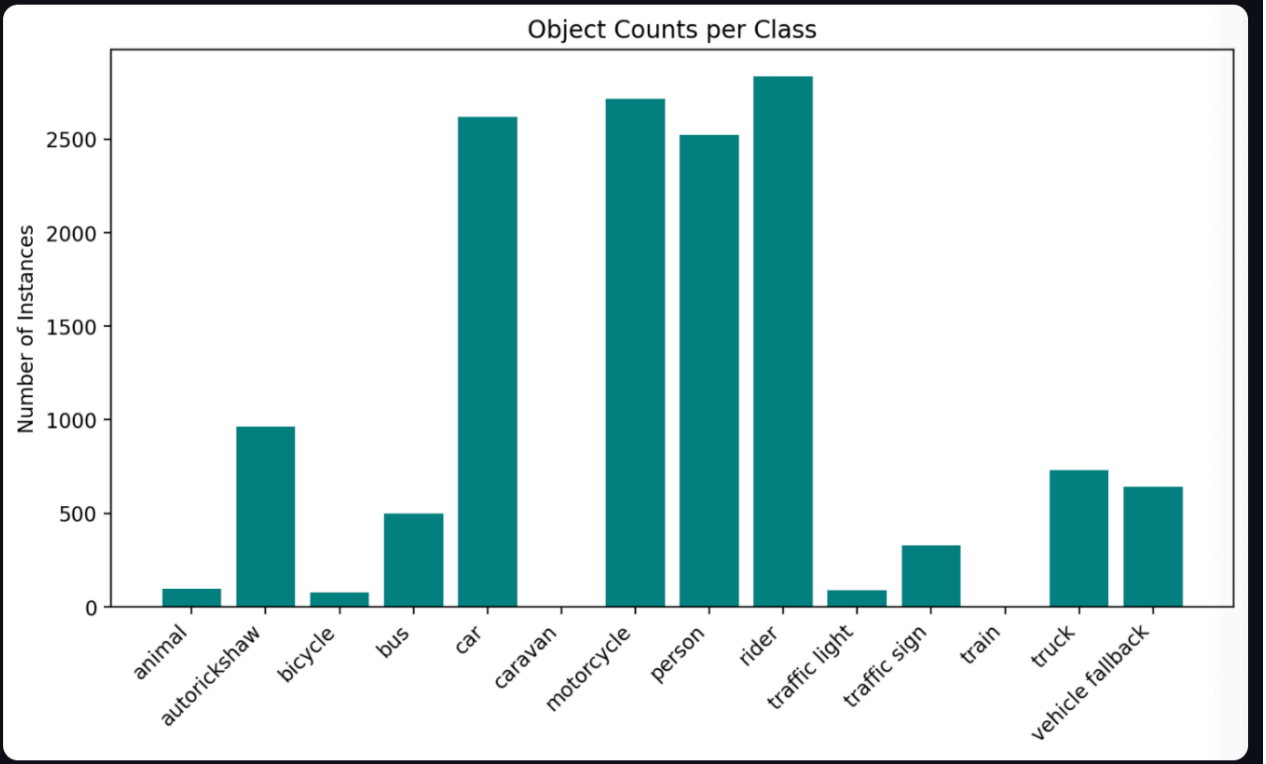
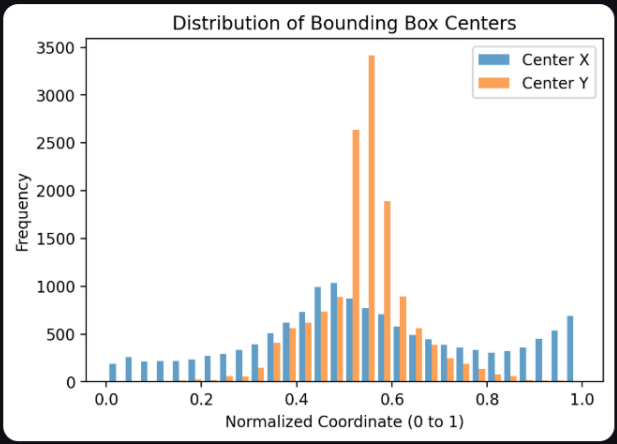


Image and Bounding Box Analysis

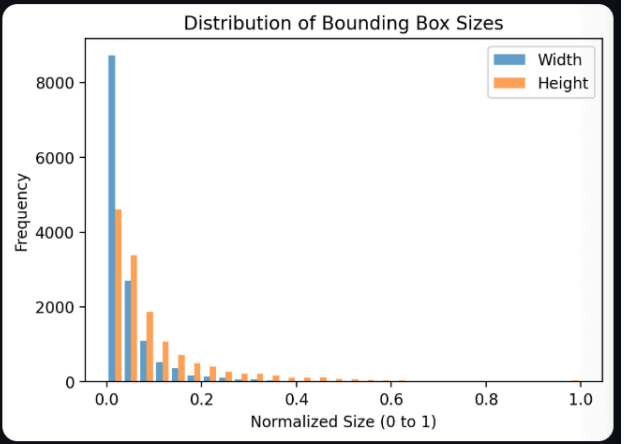
Image Dimensions: Max W: `1921` px, Max H: `1080` px. Avg W: `1870` px, Avg H: `1071` px.

Bounding Box Center Distribution



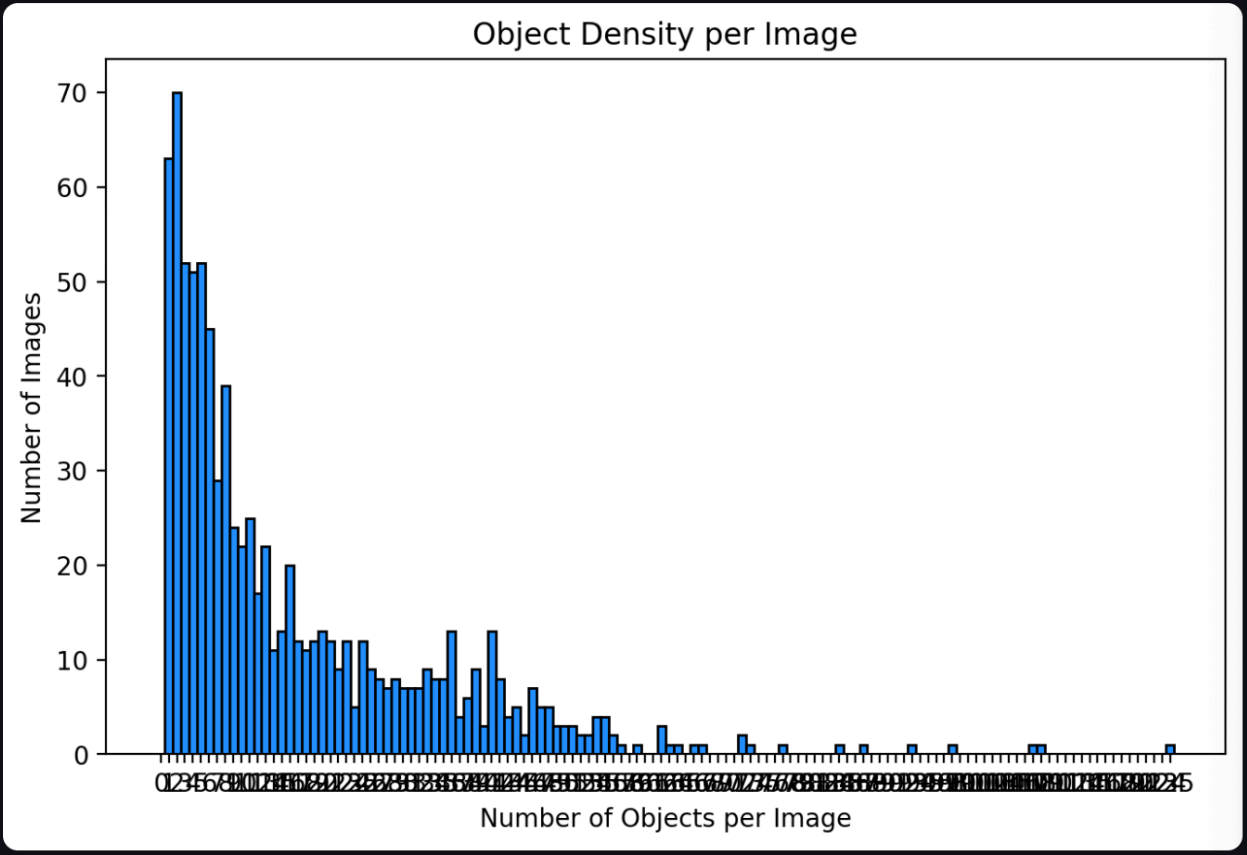
This plot reveals if objects are biased towards specific regions of the images.

Bounding Box Size Distribution



This plot helps understand the typical scale of objects in the dataset (small objects dominate if bars are near 0).

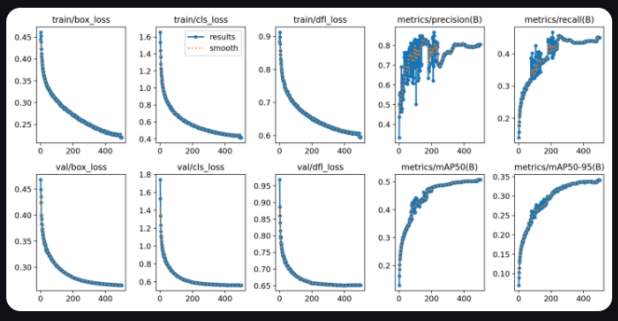
📦 Object Density Analysis



This histogram shows how many objects are typically present in each image, indicating dataset sparsity or density.

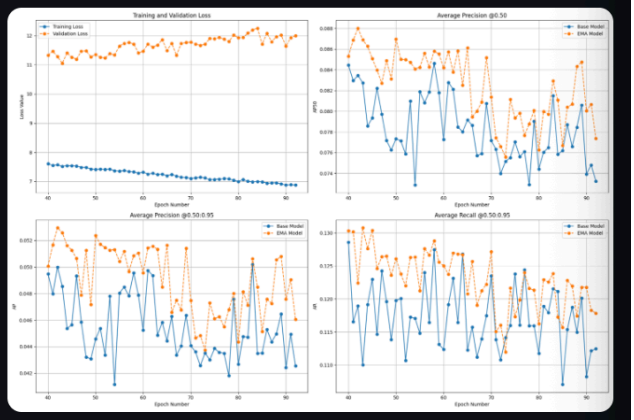
II. Model Training Visualizations

YOLOv8 Training Plots



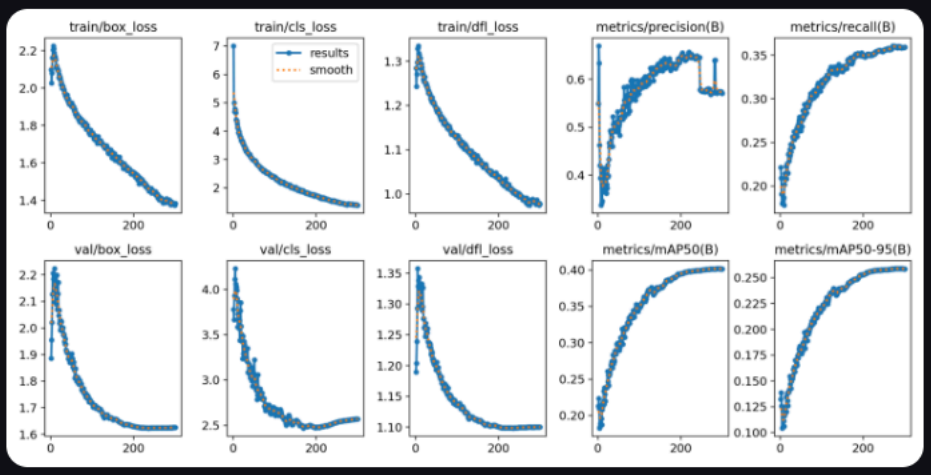
YOLOv8 Training Metrics

RF-DETR Training Plots



RF-DETR Training Metrics

YOLOv12 Training Plots



YOLOv12 Training Metrics

YOLOv8 Evaluation Results (Filtered Dataset)

⚠️ **mAP@0.50 Anomaly Detected:** Reported mAP@0.50 is anomalous or zero despite high mAP@0.50:0.95. A derived value is being used for visualization. **(Check console for original P/R values).**

⚠️ **Precision/Recall Bug:** Per-class Precision/Recall are reported as zero. Using mAP@0.50 values as a **visual proxy** for P/R in the table.

mAP@0.50	mAP@0.50:0.95
0.5887	0.4529

Class	Precision	Recall	mAP50	mAP50-95
animal	0.1892	0.1892	0.1892	0.1455
autorickshaw	0.6889	0.6889	0.6889	0.5300
bicycle	0.6585	0.6585	0.6585	0.5065
bus	0.7586	0.7586	0.7586	0.5835
car	0.6860	0.6860	0.6860	0.5277
caravan	0.9990	0.9990	0.9990	0.8955
motorcycle	0.5684	0.5684	0.5684	0.4373
person	0.4181	0.4181	0.4181	0.3216
rider	0.4903	0.4903	0.4903	0.3772
traffic light	0.4066	0.4066	0.4066	0.3128
traffic sign	0.3887	0.3887	0.3887	0.2990
trailer	0.5887	0.5887	0.5887	0.4529
train	0.9055	0.9055	0.9055	0.6965
truck	0.6986	0.6986	0.6986	0.5374
vehicle fallback	0.2206	0.2206	0.2206	0.1697

RF-DETR Evaluation Results (Validation $extmAP$)

Overall Performance Summary (Validation Set)

Overall mAP@50:95	Overall mAP@50	Overall Precision
0.0461	0.0774	0.2497

Class	mAP@50:95	mAP@50	Precision	Recall
animal	0.0141	0.0266	0.0000	0.1500
autorickshaw	0.0976	0.1498	0.5035	0.1500
bicycle	0.0141	0.0331	0.0000	0.1500
bus	0.1060	0.1550	0.5129	0.1500
car	0.0816	0.1274	0.4707	0.1500
caravan	0.0269	0.0291	0.1429	0.1500
motorcycle	0.0670	0.1227	0.4877	0.1500
person	0.0355	0.0715	0.1919	0.1500
rider	0.0637	0.1174	0.4709	0.1500
traffic light	0.0379	0.0797	0.2078	0.1500
traffic sign	0.0433	0.0827	0.2411	0.1500
trailer	0.0000	0.0000	0.0000	0.1500
train	0.0000	0.0000	0.0000	0.1500
truck	0.0875	0.1352	0.5037	0.1500
vehicle fallback	0.0157	0.0305	0.0128	0.1500
all	0.0461	0.0774	0.2497	0.1500

YOLOv12 Evaluation Results (Filtered Dataset)

⚠️ **mAP@0.50 Anomaly Detected:** Reported mAP@0.50 is anomalous or zero despite high mAP@0.50:0.95. A derived value is being used for visualization. **(Check console for original P/R values).**

⚠️ **Precision/Recall Bug:** Per-class Precision/Recall are reported as zero. Using mAP@0.50 values as a **visual proxy** for P/R in the table.

mAP@0.50	mAP@0.50:0.95
0.2895	0.2227

Class	Precision	Recall	mAP50	mAP50-95
animal	0.1697	0.1697	0.1697	0.1305
autorickshaw	0.4821	0.4821	0.4821	0.3709
bicycle	0.3963	0.3963	0.3963	0.3049
bus	0.4765	0.4765	0.4765	0.3666
car	0.4861	0.4861	0.4861	0.3739
caravan	0.0000	0.0000	0.0000	0.0000
motorcycle	0.3754	0.3754	0.3754	0.2888
person	0.2764	0.2764	0.2764	0.2126
rider	0.3464	0.3464	0.3464	0.2665
traffic light	0.2194	0.2194	0.2194	0.1688
traffic sign	0.2709	0.2709	0.2709	0.2084
trailer	0.2895	0.2895	0.2895	0.2227
train	0.0000	0.0000	0.0000	0.0000
truck	0.4474	0.4474	0.4474	0.3441
vehicle fallback	0.1062	0.1062	0.1062	0.0817